Wave Optics

Complete Conceptual Foundation, Huygens' Principle, Interference, Diffraction, Polarisation, Worked Examples & Standard Exam Practice

1. Tutorial Objectives & Exam Scope

In National Eligibility cum Entrance Test (NEET UG), Wave Optics is a vital core topic contributing 2 to 3 direct questions every year. This tutorial comprehensively addresses the entire theoretical framework, mathematical rigor, and qualitative concepts specified in the syllabus.

Key Learning Outcomes:

- Understand the historical transition from the corpuscular theory (Descartes, Newton) to the wave theory (Huygens, Young, Maxwell).
- Master the definition, geometric shapes, and properties of wavefronts and secondary wavelets.
- Derive the laws of reflection and refraction using Huygens' principle, and prove that frequency remains invariant across media while speed and wavelength vary.
- Understand optical path equivalence and wavefront modifications produced by thin prisms, convex lenses, and concave mirrors.
- Differentiate between coherent and incoherent addition of waves, deriving resultant displacement, phase-difference conditions, and intensity distributions.
- Analyze Young's Double-Slit Experiment (YDSE): fringe width formula, bright/dark fringe coordinates, and intensity variation.
- Evaluate single-slit diffraction: condition for minima and secondary maxima, central maximum width, and physical distinction from interference.
- Explain wave polarisation: transverse wave nature, plane polarized light, transmission through polaroids, and Malus' law.

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2. Historical Background & Nature of Light

The understanding of the fundamental nature of light developed through competing scientific frameworks over three centuries:

- **Corpuscular Model (Descartes & Newton):** In 1637, René Descartes proposed that light consists of tiny particles (corpuscles) and derived Snell's law. Isaac Newton elaborated this in his work *Opticks*. The corpuscular model predicted that when light refracts into an optically denser medium (bending towards the normal), its speed in the second medium must be *greater* than in the first ($v_2 > v_1$).
- **Wave Model (Christiaan Huygens, 1678):** Huygens argued that light propagates as waves spreading through a medium. His theory predicted that when a light wave refracts and bends towards the normal, its speed in the second medium must be *less* than in the first ($v_2 < v_1$).
- **Experimental Vindication (Foucault, 1850):** Jean Foucault directly measured the speed of light in water and established that the speed of light in water is strictly less than that in air ($v_{\text{water}} < v_{\text{air}}$), decisively disproving the corpuscular hypothesis and confirming the wave theory.
- **Thomas Young's Interference Experiment (1801):** Demonstrated interference using two pinholes/slits, providing indisputable physical evidence that light is a wave phenomenon and measuring visible wavelengths (e.g., yellow light $\lambda \approx 0.6\,\mu\text{m} = 600\,\text{nm}$).
- **Electromagnetic Theory (James Clerk Maxwell, ~1855):** Maxwell unified electricity and magnetism into four fundamental equations. He showed theoretically that time-varying electric and magnetic fields

regenerate each other and propagate in free space with speed $c = 1/\sqrt{\mu_0 \varepsilon_0} \approx 3 \times 10^8\,\mathrm{m/s}$. This proved that light is an electromagnetic wave requiring no mechanical medium. Heinrich Hertz produced these waves experimentally around 1890, and practical applications were pioneered by J.C. Bose and G. Marconi.

Geometrical Optics vs. Wave Optics

The wavelength of visible light is extremely small (~ 400 to $700\,\mathrm{nm}$) compared to everyday macroscopic objects, mirrors, and lenses. In the formal mathematical limit $\lambda \to 0$, the wave nature is hidden, and light energy may be regarded as traveling along straight paths called **rays**. Ray optics is therefore the asymptotic limit of wave optics when obstacle dimensions are vastly larger than λ.

3. Huygens' Principle & Wavefronts

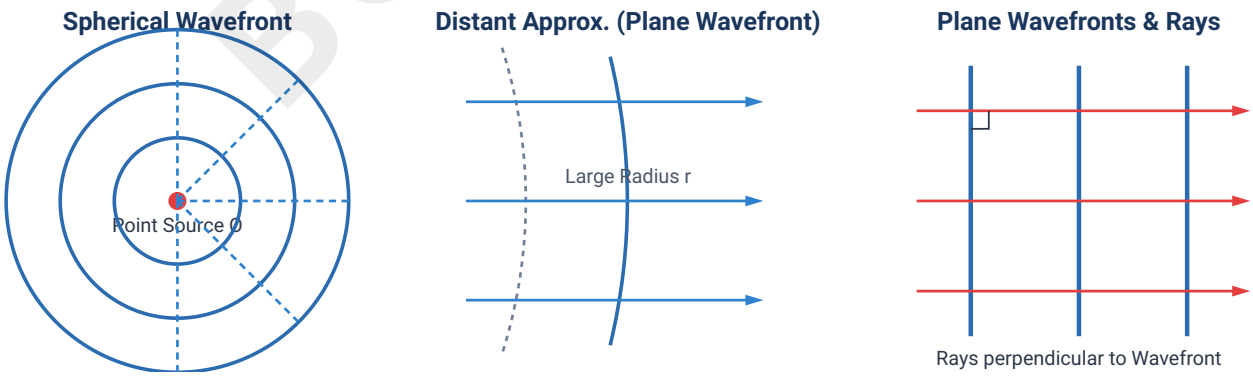
3.1 Wavefront: Definition and Geometrical Forms

When a stone is dropped into a calm pond, concentric circular ripples spread outward. All points along a given circular crest oscillate with identical phase because they are equidistant from the disturbance center.

Definition of Wavefront: A *wavefront* is defined as the locus of all adjacent points in a medium that oscillate with the same phase (i.e., a surface of constant phase).

The speed at which the wavefront travels outward is the speed of the wave v. The direction of energy propagation—the **ray of light**—is always strictly perpendicular (normal) to the wavefront at every point.

Source Configuration	Wavefront Geometry	Ray Direction	Amplitude Dependence (A)	Intensity Dependence (I)
Isotropic Point Source	Spherical surfaces centered at source	Radial outward lines	$A \propto \frac{1}{r}$	$I \propto \frac{1}{r^2}$
Linear Slit / Line Source	Coaxial Cylindrical surfaces	Radial lines in transverse planes	$A \propto \frac{1}{\sqrt{r}}$	$I \propto \frac{1}{r}$
Distant Source (Star / Sun)	Plane surfaces (parallel sheets)	Parallel straight beam	$A = \text{constant}$	$I = \text{constant}$



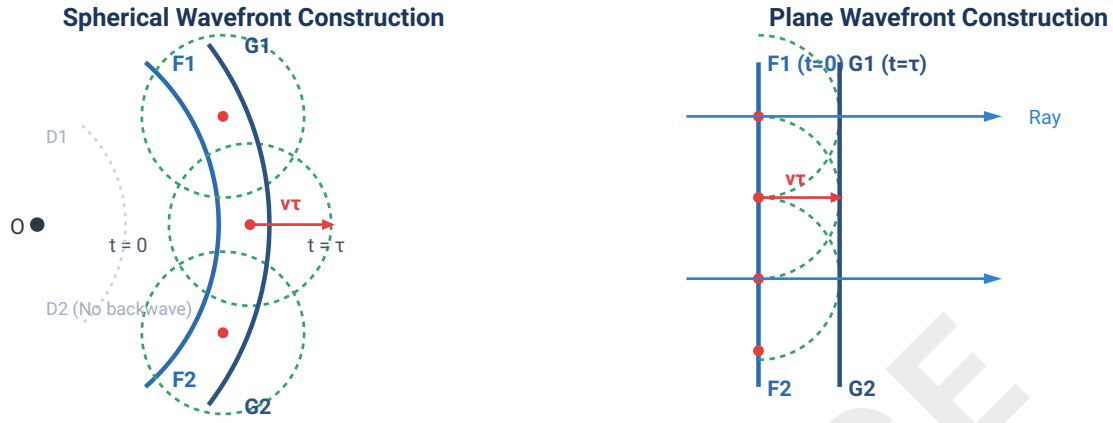
Tutorial Diagram 1: Geometrical propagation of spherical wavefronts, asymptotic transition at large distance, and parallel rays perpendicular to plane wavefronts.

3.2 Huygens' Geometrical Construction

Given the location and shape of a primary wavefront at time $t = 0$, Huygens' principle enables us to determine the position of the wavefront at a subsequent time $t = \tau$:

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1. Every point on a primary wavefront acts as a fresh source of a secondary disturbance, sending out spherical **secondary wavelets** in all directions with the characteristic phase speed v in that medium.
2. At time τ, each secondary wavelet has a radius equal to $v\tau$.
3. The forward envelope (the tangential surface touching all these secondary spherical wavelets in the forward direction) defines the new primary wavefront at $t = \tau$.



Tutorial Diagram 2: Secondary wavelets of radius $v\tau$ and their forward tangential envelopes generating subsequent wavefronts.

Absence of the Backwave

A spherical wavelet mathematically spreads outward in all 4π steradians, which geometrically permits a backward tangent envelope (D_1D_2). Huygens made the ad-hoc assumption that the amplitude of secondary wavelets is maximum in the forward direction and zero in the backward direction. Later, rigorous wave theory (formulated by Fresnel and Kirchhoff) established the obliquity (inclination) factor:

$$F(\theta) = \frac{1 + \cos\theta}{2}$$

Where θ is the angle with the forward normal. For the direct forward direction ($\theta = 0^\circ$), $F(0^\circ) = 1$ (maximum); while for the backward direction ($\theta = 180^\circ$), $F(180^\circ) = 0$, completely eliminating the backwave.

4. Refraction & Reflection of Plane Waves via Huygens' Principle

4.1 Refraction of a Plane Wave at a Denser Medium ($v_2 < v_1$)

Consider an interface PP' separating medium 1 (phase speed v_1) from medium 2 (phase speed v_2). Let an incident plane wavefront AB approach the interface at an angle of incidence i. At the instant when edge A contacts the surface, edge B has yet to travel a distance BC in medium 1 to touch the interface at C.

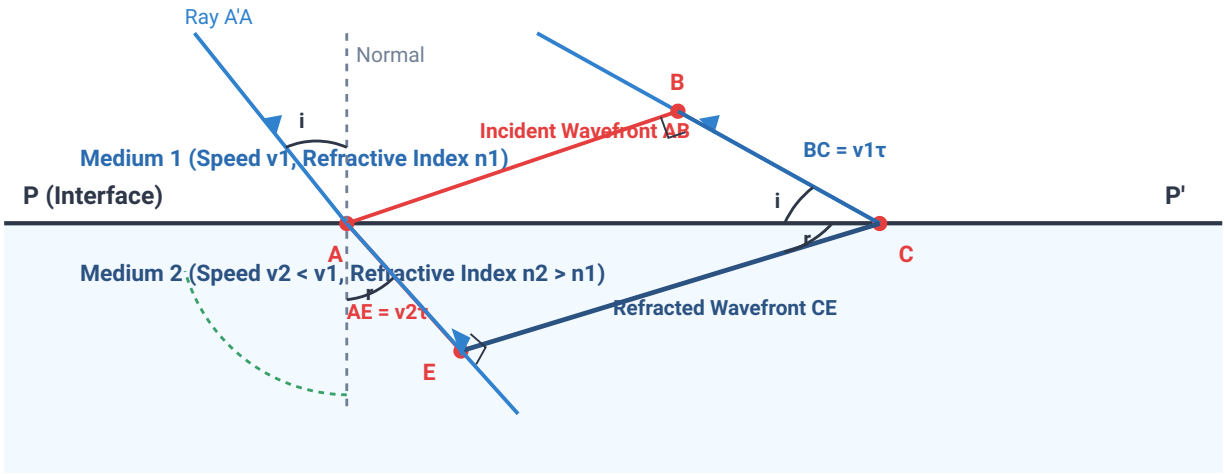
$$BC = v_1\tau$$

During the same time interval τ, the secondary disturbance originating from point A spreads into medium 2 as a hemisphere of radius:

$$AE = v_2\tau$$

Drawing the tangent line from C to this wavelet sphere in medium 2 identifies the refracted wavefront CE. The normal to CE is the refracted ray, making an angle of refraction r with the normal to PP'.

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Tutorial Diagram 3: Huygens construction for refraction of a plane wave at a denser medium boundary ($v_2 < v_1$), proving Snell's Law.

Mathematical Derivation of Snell's Law:

In right-angled triangle $\triangle ABC$:

$$\sin i = \frac{BC}{AC} = \frac{v_1 \tau}{AC}$$

In right-angled triangle $\triangle AEC$:

$$\sin r = \frac{AE}{AC} = \frac{v_2 \tau}{AC}$$

Dividing the two trigonometric ratios eliminates the common hypotenuse AC:

$$\frac{\sin i}{\sin r} = \frac{v_1}{v_2}$$

Defining the absolute refractive indices of media 1 and 2 as $n_1 = c/v_1$ and $n_2 = c/v_2$, we obtain:

$$\frac{\sin i}{\sin r} = \frac{c/n_1}{c/n_2} = \frac{n_2}{n_1} \implies n_1 \sin i = n_2 \sin r \quad \text{(Snell's Law)}$$

If $r < i$ (ray bends towards the normal), it follows unambiguously that $v_2 < v_1$. The speed of light is strictly lower in an optically denser medium.

Wavelength and Frequency Invariance:

If the distance BC contains exactly one full wavelength in medium 1 ($BC = \lambda_1$), the corresponding disturbance traveled from A to E in identical time τ is one full wavelength in medium 2 ($AE = \lambda_2$):

$$\frac{\lambda_1}{\lambda_2} = \frac{BC}{AE} = \frac{v_1 \tau}{v_2 \tau} = \frac{v_1}{v_2} \implies \frac{v_1}{\lambda_1} = \frac{v_2}{\lambda_2} = \nu$$

Fundamental Rule of Optical Interfaces:

When light crosses an interface into a denser medium ($n_2 > n_1$):

- **Frequency ν remains unchanged:** Frequency is an intrinsic property of the emitting source oscillator.
- **Speed decreases:** $v = c/n$.

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- **Wavelength decreases:** $\lambda = \lambda_0/n$.

4.2 Refraction into a Rarer Medium & Total Internal Reflection (TIR)

When light travels from an optically denser medium (n_1) to a rarer medium ($n_2 < n_1$), $v_2 > v_1$, and the refracted ray bends away from the normal ($r > i$).

As the incidence angle increases, the angle of refraction reaches 90° at a unique angle of incidence called the **critical angle** i_c:

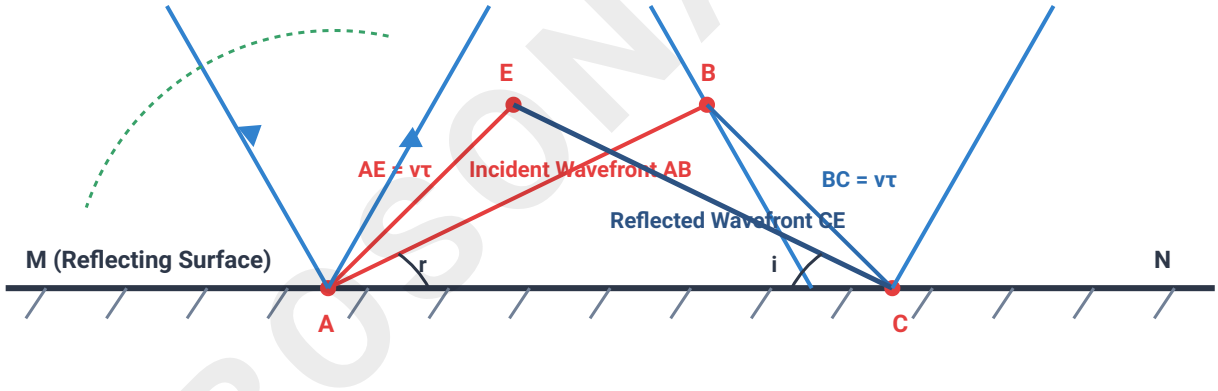
$$n_1 \sin i_c = n_2 \sin 90^\circ \implies \sin i_c = \frac{n_2}{n_1} \quad \left(\text{for air as second medium: } \sin i_c = \frac{1}{n}\right)$$

For any angle of incidence $i > i_c$, no real angle of refraction r can satisfy $\sin r \leq 1$. Consequently, no refracted wavefront can form, and the incident wave is completely reflected back into the denser medium—the phenomenon of **Total Internal Reflection (TIR)**.

4.3 Reflection of a Plane Wave at a Plane Surface

Consider a plane wave AB incident on a reflecting surface MN at angle i. The edge B reaches point C on the mirror after time τ, covering distance $BC = v\tau$. During time τ, the secondary wavelet from point A expands into the original medium as a hemisphere of radius $AE = v\tau$.

The tangent plane CE from point C forms the reflected wavefront.



Tutorial Diagram 4: Huygens geometrical construction for the reflection of a plane wave at a plane surface, proving $i = r$.

Comparing triangles $\triangle EAC$ and $\triangle BAC$:

- $\angle AEC = \angle ABC = 90^\circ$
- Hypotenuse AC is common.
- $AE = BC = v\tau$

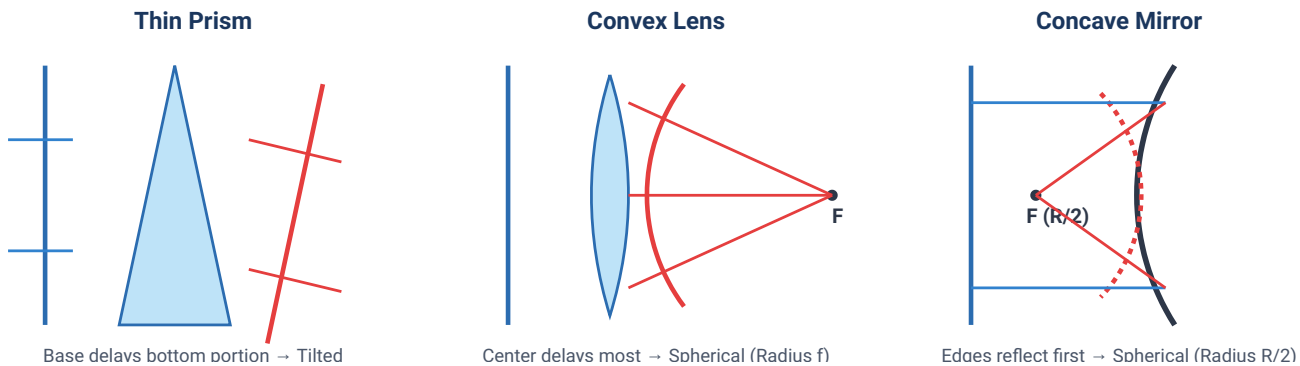
Therefore, by RHS criteria, $\triangle EAC \cong \triangle BAC$. Corresponding angles are strictly equal:

$$\angle i = \angle r \quad \text{(Law of Reflection)}$$

4.4 Wavefront Action by Prisms, Lenses, and Mirrors

The wavefront perspective provides intuitive insight into how optical components manipulate light:

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Tutorial Diagram 5: Evolution of plane wavefronts passing through a prism, a convex lens, and reflecting off a concave mirror.

Theorem of Equal Optical Path Time: The total time taken by light to travel from any point on an object wavefront to the corresponding point on the image wavefront is *strictly identical* along all physical ray trajectories. In a convex lens, although axial rays traverse a shorter geometrical distance to the focus, they traverse the thickest glass segment where speed is lowest, perfectly synchronizing arrival times with peripheral rays traveling through air.

5. Coherence and the Principle of Superposition

5.1 The Principle of Superposition

When two or more light waves propagate simultaneously through a medium, the resultant displacement $\vec{y}$ at any spatial position and time is the vector sum of individual displacements produced by each independent wave:

$$\vec{y} = \vec{y}_1 + \vec{y}_2 + \cdots + \vec{y}_n$$

5.2 Mathematical Formulation of Interference

Consider two monochromatic harmonic wave sources S_1 and S_2 vibrating in phase at frequency ω. Let their displacements at an arbitrary observation point P be:

$$y_1 = a_1 \cos(\omega t), \quad y_2 = a_2 \cos(\omega t + \phi)$$

Where ϕ is the stationary phase difference at point P arising from the spatial optical path difference $\Delta x = S_2P - S_1P$:

$$\phi = \frac{2\pi}{\lambda}\Delta x = \frac{2\pi}{\lambda}(S_2P - S_1P)$$

By superposition, the resultant displacement is:

$y = y_1 + y_2 = a_1 \cos \omega t + a_2[\cos \omega t \cos \phi - \sin \omega t \sin \phi] = (a_1 + a_2 \cos \phi) \cos \omega t - (a_2 \sin \phi) \sin \omega t$

Setting $R \cos \theta = a_1 + a_2 \cos \phi$ and $R \sin \theta = a_2 \sin \phi$, we find:

$$y = R \cos(\omega t + \theta)$$

Squaring and adding yields the resultant amplitude squared:

$$R^2 = a_1^2 + a_2^2 + 2a_1a_2 \cos \phi$$

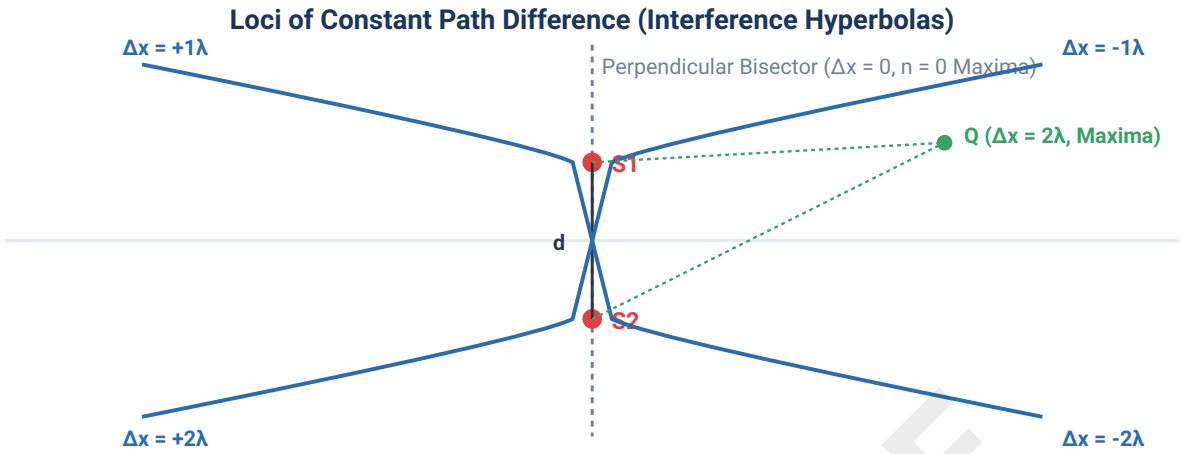
Since optical intensity is proportional to the square of amplitude ($I \propto R^2$):

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi$$

For identical source amplitudes ($a_1 = a_2 = a \implies I_1 = I_2 = I_0$):

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$$I = 2I_0(1 + \cos\phi) = 4I_0 \cos^2\left(\frac{\phi}{2}\right)$$



Tutorial Diagram 6: The geometric locus of points with constant path difference in space is a family of hyperboloids of revolution.

5.3 Conditions for Maxima and Minima

Characteristic	Constructive Interference (Bright Fringe / Antinode)	Destructive Interference (Dark Fringe / Node)
Phase Difference (ϕ)	$\phi = 0, \pm 2\pi, \pm 4\pi, \cdots = 2n\pi$	$\phi = \pm\pi, \pm 3\pi, \cdots = (2n+1)\pi$
Path Difference (Δx)	$\Delta x = n\lambda \quad (n = 0, 1, 2, \ldots)$	$\Delta x = \left(n + \frac{1}{2}\right)\lambda \quad (n = 0, 1, 2, \ldots)$
Resultant Amplitude	$A_{\max} = a_1 + a_2 \quad$ (if equal: $2a$)	$A_{\min} = \lvert a_1 - a_2 \rvert \quad$ (if equal: 0)
Resultant Intensity	$I_{\max} = (\sqrt{I_1} + \sqrt{I_2})^2 \quad$ (if equal: $4I_0$)	$I_{\min} = (\sqrt{I_1} - \sqrt{I_2})^2 \quad$ (if equal: 0)

5.4 Coherent vs. Incoherent Superposition

- **Coherent Sources:** Two sources maintaining a time-invariant phase difference ϕ. They produce a stable, stationary interference pattern where positions of maxima and minima do not fluctuate.
- **Incoherent Sources (Two Independent Light Bulbs / Sodium Lamps):** Ordinary thermal sources emit light through atomic transitions lasting approximately 10^{-10} s. In independent sources, phase changes occur randomly every 10^{-10} s. Detectors (and the human eye, persistence of vision ≈ 0.1 s) register the time-average of the interference term:

  $$\langle \cos\phi \rangle_{\text{time}} = 0$$

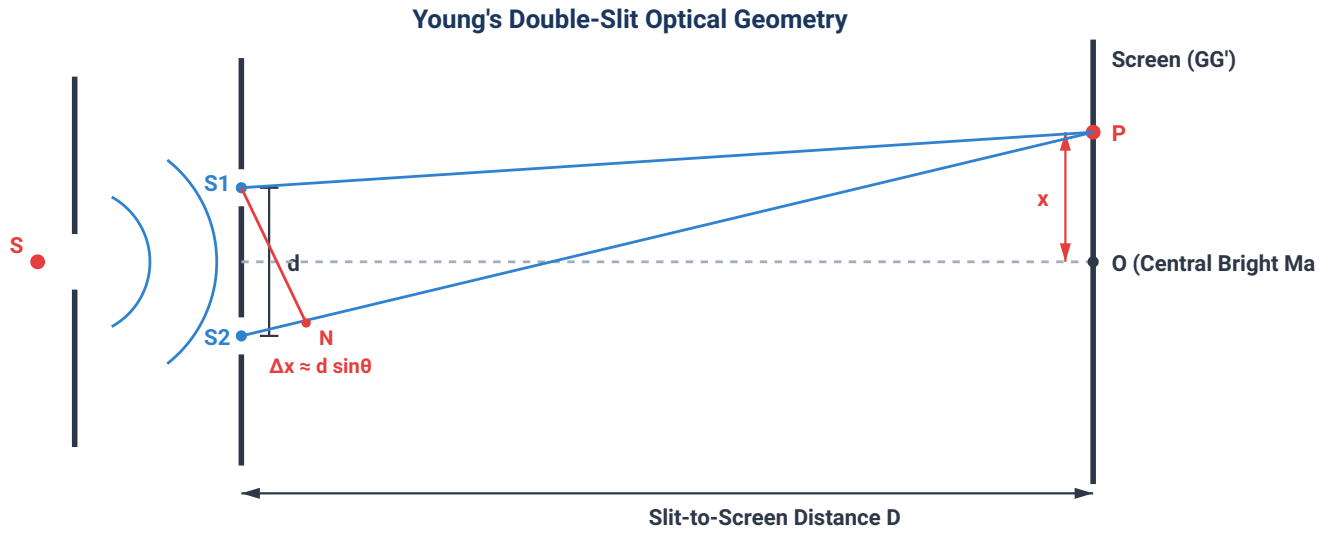
  The resulting intensity is a uniform, featureless illumination:

  $$I = I_1 + I_2 = 2I_0 \quad \text{(Direct scalar intensity addition)}$$

6. Young's Double-Slit Experiment (YDSE)

Thomas Young solved the coherence challenge by using a single primary pinhole/slit S to illuminate two secondary pinholes S_1 and S_2. Because both S_1 and S_2 receive wavefront segments derived from the *same original wavefront*, any random phase jump in S transfers identically and simultaneously to both, locking their relative phase difference permanently to zero ($\phi_0 = 0$).

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Tutorial Diagram 7: Young's double-slit experiment arrangement showing geometric path difference $\Delta x = S_2P - S_1P \approx xd/D$.

6.1 Derivation of Path Difference and Fringe Width

Using the coordinate geometry of the apparatus where $S_1 = (0, +d/2)$, $S_2 = (0, -d/2)$, and screen is at $z = D$, the distances to point $P(x)$ are:

$$(S_2P)^2 = D^2 + \left(x + \frac{d}{2}\right)^2, \quad (S_1P)^2 = D^2 + \left(x - \frac{d}{2}\right)^2$$

Subtracting the two equations:

$$(S_2P)^2 - (S_1P)^2 = (S_2P - S_1P)(S_2P + S_1P) = 4 \cdot x \cdot \frac{d}{2} = 2xd$$

In standard laboratory experiments, the slit spacing is minuscule compared to screen distance ($d \ll D$) and fringe positions are paraxial ($x \ll D$). Hence, $S_2P + S_1P \approx 2D$:

$$\Delta x = S_2P - S_1P \approx \frac{2xd}{2D} = \frac{xd}{D} = d\sin\theta \approx d\theta$$

Position of Bright Fringes (Maxima):

$$\frac{x_n d}{D} = n\lambda \implies x_n = \frac{n\lambda D}{d}, \quad n = 0, \pm 1, \pm 2, \ldots$$

Position of Dark Fringes (Minima):

$$\frac{x_n d}{D} = \left(n + \frac{1}{2}\right)\lambda \implies x'_n = \left(n + \frac{1}{2}\right)\frac{\lambda D}{d}, \quad n = 0, \pm 1, \pm 2, \ldots$$

Fringe Width (β):

The fringe width β is defined as the linear distance separating any two consecutive bright or two consecutive dark fringes:

$$\beta = x_{n+1} - x_n = \frac{(n+1)\lambda D}{d} - \frac{n\lambda D}{d} = \frac{\lambda D}{d}$$

Angular Fringe Width (θ_0):

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$$\theta_0 = \frac{\beta}{D} = \frac{\lambda}{d} \quad (\text{independent of screen distance } D)$$

Impact of Medium Immersion:

If the entire YDSE apparatus is submerged into a liquid of refractive index n, the wavelength contracts to $\lambda' = \lambda/n$, while D and d remain fixed:

$$\beta' = \frac{\lambda' D}{d} = \frac{\beta}{n}$$

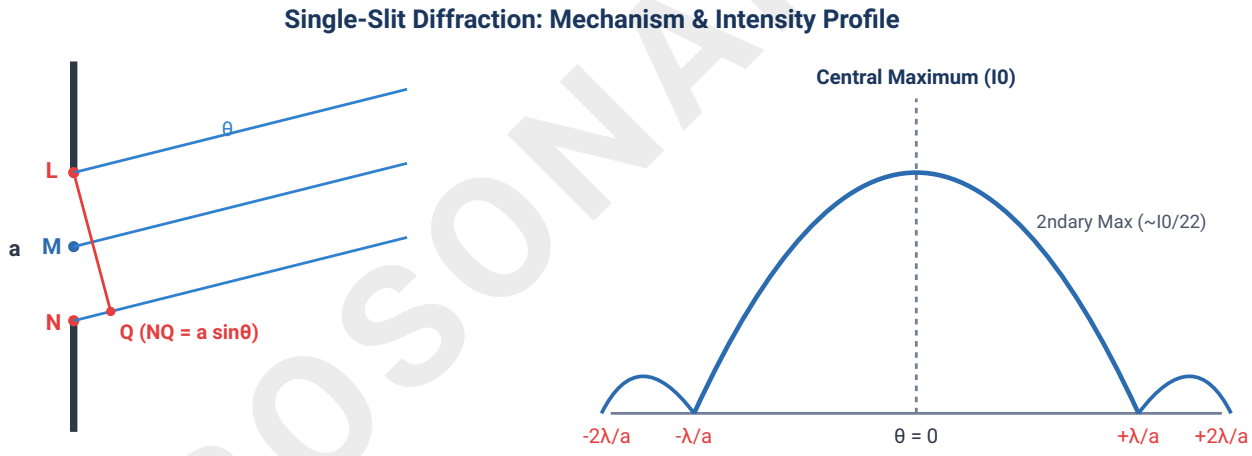
The interference fringe pattern shrinks uniformly by a factor of n.

7. Single-Slit Fraunhofer Diffraction

Diffraction is the bending or spreading of waves around obstacles and the edges of apertures into regions of geometrical shadow. It occurs when obstacle/slit dimensions are comparable to the wavelength of light ($a \sim \lambda$).

7.1 Single Slit Analysis

A plane wavefront of monochromatic light (λ) falls normally on a narrow slit LN of width a. The diffracted light is collected on a distant screen at distance D.



Tutorial Diagram 8: Single-slit diffraction geometry showing slit subdivision and the central maximum flanked by decaying secondary maxima.

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Condition for Minima (Zero Intensity):

To find the angle θ at which intensity drops to zero, divide the slit LN into two equal halves LM and MN, each of width $a/2$. For every point in LM, there is a corresponding point in MN at distance $a/2$ whose path difference to the screen is:

$$\Delta x = \frac{a}{2}\sin\theta$$

When this path difference equals $\lambda/2$, the wavelet pairs arrive in complete opposite phase (π) and destructively cancel out everywhere. Setting $\frac{a}{2}\sin\theta = \frac{\lambda}{2}$ gives $\sin\theta = \lambda/a$.

Dividing the slit into $2n$ equal sub-zones generalizes the condition for all diffraction minima:

$$a \sin\theta = n\lambda \implies \theta \approx \frac{n\lambda}{a}, \quad n = \pm 1, \pm 2, \pm 3, \ldots$$

Condition for Secondary Maxima:

Dividing the slit into $(2n+1)$ equal zones leaves one uncancelled zone that transmits net energy to the screen:

$$\theta \approx \left(n + \frac{1}{2}\right)\frac{\lambda}{a}, \quad n = \pm 1, \pm 2, \pm 3, \ldots$$

Width of the Central Diffraction Maximum:

The central bright maximum spans between the first minimum on either side ($-\lambda/a$ to $+\lambda/a$):

Angular Width: $2\theta_1 = \frac{2\lambda}{a}$
Linear Width on Screen: $\beta_0 = 2x_1 = \frac{2\lambda D}{a}$

Notice that the central diffraction maximum is **twice as wide** as any secondary maximum ($\lambda D/a$).

7.2 Interference vs. Diffraction Comparison

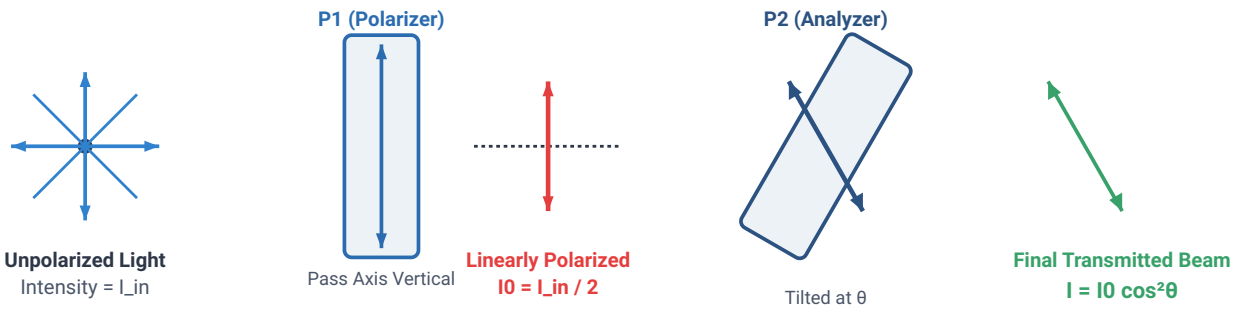
Characteristic	Interference (Two Slits)	Diffraction (Single Slit)
Origin	Superposition of waves from two distinct coherent wavefront segments	Superposition of infinite secondary wavelets originating from continuous points of the *same* slit
Fringe Width	All fringes (central and higher order) have identical width $\beta = \lambda D/d$	Central maximum has width $2\lambda D/a$, twice as wide as secondary bands
Intensity Distribution	All bright fringes possess equal intensity ($I_{\max} = 4I_0$)	Intensity decays sharply from central peak: central (I_0), 1st secondary ($\approx I_0/22$), 2nd secondary ($\approx I_0/61$)
Minimum Condition	$d\sin\theta = (n+1/2)\lambda$	$a\sin\theta = n\lambda$
Conservation of Energy	Strictly conserved; energy is redistributed from dark zones to bright zones	Strictly conserved; energy redistributed into diffraction lobes

8. Polarisation of Light

8.1 Transverse Wave Nature

Interference and diffraction are exhibited by all waves, including longitudinal acoustic waves. In contrast, **polarisation** is exhibited exclusively by **transverse waves**, proving that light is a transverse electromagnetic wave where the electric vector $\vec{E}$ oscillates perpendicularly to the propagation direction $\vec{k}$.

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Tutorial Diagram 9: Polarization of unpolarized light by Polaroid P_1 and intensity modulation through Analyzer P_2 obeying Malus' Law.

8.2 Polaroids and Pass-Axis

A polaroid sheet consists of long-chain polymer molecules (such as polyvinyl alcohol treated with iodine) aligned strictly parallel along a manufacturing axis. Conduction electrons slide freely along the aligned chain axis, dissipating the electric field component parallel to the molecules via Joule heating. The electric field component perpendicular to the chains traverses the sheet unimpeded. This perpendicular direction is termed the **pass-axis**.

- When natural (unpolarized) light of intensity I_{in} enters a polaroid P_1, the time average $\langle \cos^2 \theta \rangle = 1/2$ over all random orientations transmits exactly half the incident intensity:

$$I_0 = \frac{1}{2} I_{\text{in}}$$

- Rotating a single polaroid P_1 in front of an unpolarized lamp causes **no change** in transmitted intensity.

8.3 Malus' Law

When linearly polarized light of amplitude E_0 and intensity I_0 is incident upon a second polaroid sheet (analyzer P_2) whose pass-axis is oriented at an angle θ relative to the polarization direction of the incident wave, only the component along the analyzer pass-axis is transmitted:

$$E = E_0 \cos \theta$$

Squaring both sides gives **Malus' Law**:

$$I = I_0 \cos^2 \theta$$

- **Parallel Polaroids ($\theta = 0^\circ$ or 180°):** $\cos^2(0^\circ) = 1 \implies I = I_0$ (Maximum transmission).
- **Crossed Polaroids ($\theta = 90^\circ$):** $\cos^2(90^\circ) = 0 \implies I = 0$ (Total extinction / darkness).

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9. High-Yield Worked Examples

Worked Example 1: Wavelength, Speed, and Frequency Invariance

Problem: Monochromatic light of vacuum wavelength $\lambda = 589\,\text{nm}$ is incident from air onto a still water surface. The refractive index of water is $n = 1.33 = 4/3$. Speed of light in vacuum is $c = 3.0 \times 10^8\,\text{m/s}$. Determine:

1. The frequency, wavelength, and speed of the reflected light.
2. The frequency, wavelength, and speed of the refracted light.

Complete Worked Solution:

(1) Reflected Light (Propagates back in Air, $n = 1$):

- **Speed:** $v_{\text{refl}} = c = 3.0 \times 10^8\,\text{m/s}$.
- **Wavelength:** $\lambda_{\text{refl}} = 589\,\text{nm} = 5.89 \times 10^{-7}\,\text{m}$.
- **Frequency:** $\nu = \frac{c}{\lambda} = \frac{3.0\times10^8}{5.89\times10^{-7}} \approx 5.09 \times 10^{14}\,\text{Hz}$.

(2) Refracted Light (Propagates in Water, $n = 1.33$):

- **Frequency:** Light frequency is determined by atomic oscillator frequency at the source and remains invariant:

$$\nu_{\text{refr}} = \nu = 5.09 \times 10^{14}\,\text{Hz}$$

- **Speed:**

$$v_{\text{water}} = \frac{c}{n} = \frac{3.0 \times 10^8}{1.333} = 2.25 \times 10^8\,\text{m/s}$$

- **Wavelength:**

$$\lambda_{\text{water}} = \frac{\lambda_0}{n} = \frac{589\,\text{nm}}{1.333} \approx 442.8\,\text{nm}$$

Worked Example 2: YDSE Determination of Unknown Wavelength

Problem: In a double-slit experiment, the two slits are separated by $d = 0.28\,\text{mm}$ and the screen is placed at $D = 1.4\,\text{m}$. The linear distance between the central bright fringe and the fourth bright fringe is measured to be $1.2\,\text{cm}$. Calculate the wavelength of light used.

Complete Worked Solution:

Convert all values into standard SI units:

- $d = 0.28 \times 10^{-3}\,\text{m}$
- $D = 1.4\,\text{m}$
- $x_4 = 1.2 \times 10^{-2}\,\text{m}$ for order $n = 4$

The position formula for the n-th bright fringe is:

$$x_n = \frac{n\lambda D}{d} \implies \lambda = \frac{x_n \cdot d}{n \cdot D}$$

Substitute the given quantities:

$$\lambda = \frac{(1.2 \times 10^{-2}\,\text{m}) \times (0.28 \times 10^{-3}\,\text{m})}{4 \times 1.4\,\text{m}} = \frac{3.36 \times 10^{-6}}{5.6} = 6.0 \times 10^{-7}\,\text{m} = 600\,\text{nm}$$

BOSONANCE

Worked Example 3: Intensity at a Fractional Path Difference

Problem: In Young's double-slit experiment using monochromatic light of wavelength λ, the intensity of light at a point on the screen where the path difference is λ is found to be K units. What is the intensity at a point where the path difference is $\lambda/3$?

Complete Worked Solution:

Step 1: Relate K to source maximum intensity:

A path difference of $\Delta x_1 = \lambda$ corresponds to phase difference $\phi_1 = \frac{2\pi}{\lambda}(\lambda) = 2\pi$.

$$I_1 = 4I_0 \cos^2\left(\frac{2\pi}{2}\right) = 4I_0 \cos^2(\pi) = 4I_0(1) = K \implies 4I_0 = K$$

Step 2: Calculate intensity for $\Delta x_2 = \lambda/3$:

The phase difference is:

$$\phi_2 = \frac{2\pi}{\lambda}\left(\frac{\lambda}{3}\right) = \frac{2\pi}{3} = 120^\circ$$

The intensity is:

$$I_2 = 4I_0 \cos^2\left(\frac{\phi_2}{2}\right) = K\cos^2\left(\frac{\pi}{3}\right) = K\left(\frac{1}{2}\right)^2 = \frac{K}{4}$$

Worked Example 4: Coincidence of Bright Fringes for Two Wavelengths

Problem: A light beam containing two wavelengths $\lambda_1 = 650\,\text{nm}$ and $\lambda_2 = 520\,\text{nm}$ is used in a double-slit experiment. Slit separation is $d = 2.0\,\text{mm}$ and screen distance is $D = 1.2\,\text{m}$.

1. Find the distance of the 3rd bright fringe for $\lambda_1 = 650\,\text{nm}$ from the central maximum.
2. Find the least distance from the central maximum where bright fringes from both wavelengths coincide.

Complete Worked Solution:

(1) Distance of 3rd bright fringe for $\lambda_1 = 650\,\text{nm}$:

$$x = \frac{n_1\lambda_1 D}{d} = \frac{3 \times (650 \times 10^{-9}\,\text{m}) \times 1.2\,\text{m}}{2.0 \times 10^{-3}\,\text{m}} = \frac{2.34 \times 10^{-6}}{2.0 \times 10^{-3}} = 1.17 \times 10^{-3}\,\text{m} = 1.17\,\text{mm}$$

(2) Smallest coincidence position:

Let the n_1-th bright fringe of λ_1 coincide with the n_2-th bright fringe of λ_2:

$$x = \frac{n_1\lambda_1 D}{d} = \frac{n_2\lambda_2 D}{d} \implies n_1\lambda_1 = n_2\lambda_2$$

$$\frac{n_1}{n_2} = \frac{\lambda_2}{\lambda_1} = \frac{520\,\text{nm}}{650\,\text{nm}} = \frac{4}{5}$$

Since n_1 and n_2 must be positive integers, the smallest integer values are $n_1 = 4$ and $n_2 = 5$. The 4th bright fringe of $650\,\text{nm}$ coincides with the 5th bright fringe of $520\,\text{nm}$:

$$x_{\min} = \frac{4 \times (650 \times 10^{-9}) \times 1.2}{2.0 \times 10^{-3}} = \frac{3.12 \times 10^{-6}}{2.0 \times 10^{-3}} = 1.56 \times 10^{-3}\,\text{m} = 1.56\,\text{mm}$$

BOSONANCE

Worked Example 5: Rotating a Polaroid between Crossed Polaroids

Problem: Two polaroids P_1 and P_3 are set up in a crossed orientation such that no light is transmitted. A third polaroid P_2 is inserted between them and rotated. If linearly polarized light of intensity I_0 emerges from P_1, derive the intensity transmitted by P_3 as a function of the angle θ between the pass-axes of P_1 and P_2. At what angle is transmission maximized?

Complete Worked Solution:

Let the pass-axis of P_1 be along the vertical reference (0°). Since P_1 and P_3 are crossed, the pass-axis of P_3 is at 90° ($\pi/2$).

Let the pass-axis of P_2 make an angle θ with P_1. The intensity emerging from P_2 by Malus' Law is:

$$I_2 = I_0 \cos^2\theta$$

The light emerging from P_2 is polarized along its pass-axis θ. The angle between the pass-axis of P_2 and the pass-axis of P_3 is:

$$\alpha = \frac{\pi}{2} - \theta$$

Applying Malus' Law at the third polaroid P_3:

$$I_3 = I_2 \cos^2\left(\frac{\pi}{2} - \theta\right) = (I_0 \cos^2\theta)(\sin^2\theta)$$

Using trigonometric identity $\sin 2\theta = 2\sin\theta\cos\theta \implies \cos\theta\sin\theta = \frac{1}{2}\sin 2\theta$:

$$I_3 = I_0(\cos\theta\sin\theta)^2 = I_0\left(\frac{1}{2}\sin 2\theta\right)^2 = \frac{I_0}{4}\sin^2(2\theta)$$

The transmitted intensity reaches its maximum when $\sin^2(2\theta) = 1$, which occurs at $2\theta = 90^\circ \implies \theta = 45^\circ$ ($\pi/4$).

$$I_{3,\max} = \frac{I_0}{4}$$

10. NEET UG Exam-Aligned Practice Questions

Practice Question 1

Wavefront & Speed

Which of the following historical predictions and experimental observations concerning the speed of light in a denser medium is correct?

(A) Newton predicted $v_{\text{water}} < v_{\text{air}}$, confirmed by Foucault

(B) Descartes predicted $v_{\text{water}} > v_{\text{air}}$, refuted by Foucault

(C) Huygens predicted $v_{\text{water}} > v_{\text{air}}$, confirmed by Young

(D) Maxwell predicted $v_{\text{water}} = v_{\text{air}}$, refuted by Hertz

Correct Answer: (B)

Explanation: The corpuscular model of Descartes and Newton asserted that light particles accelerate upon crossing into a denser medium when bending towards the normal, predicting $v_{\text{denser}} > v_{\text{rarer}}$. In 1850, Foucault experimentally measured the speed of light in water and found it to be strictly less than in air ($v_{\text{water}} < v_{\text{air}}$), refuting the corpuscular prediction and confirming Huygens' wave theory.

BOSONANCE

Practice Question 2

Wavefront Shapes

A point source of light is placed at the principal focus of a thin convex lens. The shape of the wavefront of light emerging from the lens is:

(A) Diverging spherical

(B) Converging spherical

(C) Plane

(D) Cylindrical

Correct Answer: (C)

Explanation: When a point source is situated at the principal focus of a convex lens, the refracted rays emerge parallel to the principal axis. Rays are everywhere normal to wavefronts; therefore, a beam of parallel rays corresponds to a **plane wavefront**.

Practice Question 3

Incoherent Sources

Two separate, independent sodium lamps illuminate two closely spaced pinholes. The pattern observed on a distant viewing screen is:

(A) Sharp, stationary bright and dark interference fringes

(B) Uniform, continuous illumination with no fringes

(C) A central bright fringe surrounded by dark bands

(D) Linearly polarized light fringes

Correct Answer: (B)

Explanation: Independent light sources undergo uncorrelated phase jumps on time scales of $\sim 10^{-10}\,\mathrm{s}$. Because they are incoherent, the time-averaged interference term $\langle \cos\phi \rangle = 0$. The intensities add directly ($I = I_1 + I_2 = 2I_0$), producing uniform illumination without observable interference fringes.

Practice Question 4

YDSE Fringe Width

In a Young's double-slit experiment, the fringe width is measured to be $\beta = 0.4\,\mathrm{mm}$. If the entire experimental setup is immersed into water of refractive index $n = 4/3$, the new fringe width will be:

(A) $0.53\,\mathrm{mm}$

(B) $0.40\,\mathrm{mm}$

(C) $0.30\,\mathrm{mm}$

(D) $0.10\,\mathrm{mm}$

Correct Answer: (C)

Explanation: Fringe width is given by $\beta = \lambda D/d$. Upon immersion in water, the wavelength changes to $\lambda' = \lambda/n$. Therefore:

$$\beta' = \frac{\beta}{n} = \frac{0.4\,\mathrm{mm}}{4/3} = 0.4 \times \frac{3}{4} = 0.30\,\mathrm{mm}$$

BOSONANCE

Practice Question 5

Intensity Ratio in YDSE

In a double-slit experiment, the ratio of maximum to minimum intensity in the interference pattern is $25:9$. The ratio of the slit widths (assumed proportional to intensities) is:

(A) $4:1$

(B) $16:1$

(C) $5:3$

(D) $25:1$

Correct Answer: (B)

Explanation:

$$\frac{I_{\max}}{I_{\min}} = \left(\frac{a_1 + a_2}{a_1 - a_2}\right)^2 = \frac{25}{9} \implies \frac{a_1 + a_2}{a_1 - a_2} = \frac{5}{3}$$

Applying componendo and dividendo:

$$\frac{2a_1}{2a_2} = \frac{5+3}{5-3} = \frac{8}{2} = 4 \implies \frac{a_1}{a_2} = 4$$

Since slit width $w \propto I \propto a^2$:

$$\frac{w_1}{w_2} = \frac{I_1}{I_2} = \left(\frac{a_1}{a_2}\right)^2 = 4^2 = 16:1$$

Practice Question 6

Diffraction Minima

Light of wavelength $600\,\text{nm}$ is incident normally on a single slit of width $0.2\,\text{mm}$. The angular spread (in radians) of the central diffraction maximum is:

(A) $3.0 \times 10^{-3}\,\text{rad}$

(B) $6.0 \times 10^{-3}\,\text{rad}$

(C) $1.2 \times 10^{-3}\,\text{rad}$

(D) $1.5 \times 10^{-3}\,\text{rad}$

Correct Answer: (B)

Explanation: The first minimum occurs at $\theta_1 = \lambda/a$. The total angular spread of the central maximum extends between the first minima on either side:

$$2\theta_1 = \frac{2\lambda}{a} = \frac{2 \times (600 \times 10^{-9}\,\text{m})}{0.2 \times 10^{-3}\,\text{m}} = \frac{1.2 \times 10^{-6}}{0.2 \times 10^{-3}} = 6.0 \times 10^{-3}\,\text{rad}$$

BOSONANCE

Practice Question 7

Malus' Law

Unpolarized light of intensity $I_{\text{in}} = 64\,\text{W/m}^2$ passes through two polaroids whose pass-axes are oriented at 60° to each other. The intensity of the light emerging from the second polaroid is:

(A) $32\,\text{W/m}^2$

(B) $16\,\text{W/m}^2$

(C) $8\,\text{W/m}^2$

(D) $4\,\text{W/m}^2$

Correct Answer: (C)

Explanation:

1. After passing through the first polaroid P_1, unpolarized light becomes linearly polarized with intensity:

$$I_0 = \frac{I_{\text{in}}}{2} = \frac{64}{2} = 32\,\text{W/m}^2$$

2. After traversing the analyzer P_2 at $\theta = 60^\circ$, Malus' Law applies:

$$I = I_0 \cos^2(60^\circ) = 32 \times \left(\frac{1}{2}\right)^2 = 32 \times \frac{1}{4} = 8\,\text{W/m}^2$$

Practice Question 8

Assertion & Reason (NEET Pattern)

Assertion (A): When monochromatic light travels from air into glass, its frequency remains unchanged while its wavelength decreases.
Reason (R): Energy carried by a light wave is determined strictly by its speed of propagation.

(A) Both (A) and (R) are true and (R) is the correct explanation of (A)

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(C) (A) is true but (R) is false

(D) Both (A) and (R) are false

Correct Answer: (C)

Explanation: Frequency is an intrinsic property determined by the atomic oscillator at the source and does not change across boundaries. In glass ($n > 1$), $v = c/n$ and $\lambda = \lambda_0/n$, so (A) is true. However, the energy carried by a light wave depends on its amplitude squared (and in photon theory on frequency $h\nu$), not on propagation speed. Thus (R) is false.

BOSONANCE

Practice Question 9

Statement-Based MCQ

Consider the following two statements regarding Huygens' principle:
Statement I: Secondary wavelets spread in all directions with the speed of the wave in that medium.
Statement II: Huygens satisfactorily justified the absence of the backwave from fundamental wave equations without any ad-hoc assumption.

(A) Statement I is correct, Statement II is incorrect

(B) Statement I is incorrect, Statement II is correct

(C) Both Statement I and Statement II are correct

(D) Both Statement I and Statement II are incorrect

Correct Answer: (A)

Explanation: Statement I is Huygens' core postulate: every point on a wavefront acts as a secondary source radiating spherical wavelets at speed v. Statement II is incorrect because Huygens' original explanation was an ad-hoc assumption that wavelet amplitude is maximum forward and zero backward. Rigorous theoretical justification was provided much later by Fresnel and Kirchhoff using wave diffraction integrals.

Practice Question 10

Match the Columns

Match the optical devices in Column I with the nature of the wavefront produced in Column II when a plane wavefront is incident normally upon them:

Column I (Optical Device)	Column II (Emergent Wavefront)
(A) Thin Prism	(1) Spherical converging to focal point F
(B) Thin Convex Lens	(2) Spherical diverging
(C) Concave Mirror	(3) Plane wavefront tilted relative to incident wavefront
(D) Concave Lens	(4) Spherical converging of radius $R/2$

(A) A-3, B-1, C-4, D-2

(B) A-1, B-3, C-2, D-4

(C) A-3, B-4, C-1, D-2

(D) A-2, B-1, C-4, D-3

Correct Answer: (A)

Explanation:

- **Thin Prism:** Greater delay at the thicker base tilts the plane wavefront (A-3).
- **Convex Lens:** Maximum delay at thick center produces a converging spherical wavefront of radius f (B-1).
- **Concave Mirror:** Reflection produces a converging spherical wavefront toward focus at $R/2$ (C-4).
- **Concave Lens:** Thicker edges delay peripheral rays, producing a diverging spherical wavefront (D-2).

BOSONANCE

Practice Question 11

Fringe Disappearance

In Young's double-slit experiment, if monochromatic light is replaced by a white light source, the observer sees:

(A) Complete darkness across the entire screen

(B) Continuous white illumination with no distinct colors

(C) A central white fringe flanked by a few colored fringes, with violet closest to center

(D) A central dark fringe with alternating rainbow fringes

Correct Answer: (C)

Explanation: At the central position $x = 0$, path difference is zero for all wavelengths ($\Delta x = 0$). All constituent spectral colors interfere constructively, forming a central white fringe. For higher orders, fringe position is proportional to wavelength ($x_n \propto \lambda$). Since $\lambda_{\text{violet}} < \lambda_{\text{red}}$, the violet fringe appears closest to the central white fringe, followed by blue, green, and red, after which fringes overlap into uniform white illumination.

Practice Question 12

Polarizer Rotation

When unpolarized light of intensity I_0 is incident on a system of two crossed polarizers P_1 and P_3, the transmitted intensity is zero. If a third polarizer P_2 is placed between them with its pass-axis oriented at 30° to the pass-axis of P_1, the final transmitted intensity is:

(A) $\frac{3}{32}I_0$

(B) $\frac{3}{16}I_0$

(C) $\frac{1}{8}I_0$

(D) $\frac{1}{16}I_0$

Correct Answer: (A)

Explanation:

1. Light transmitted through P_1: $I_1 = \frac{I_0}{2}$.
2. Light transmitted through P_2 (at $\theta = 30^\circ$):

$$I_2 = I_1 \cos^2(30^\circ) = \frac{I_0}{2} \times \left(\frac{\sqrt{3}}{2}\right)^2 = \frac{I_0}{2} \times \frac{3}{4} = \frac{3}{8}I_0$$

3. Angle between P_2 and P_3 is $90^\circ - 30^\circ = 60^\circ$. Transmitted through P_3:

$$I_3 = I_2 \cos^2(60^\circ) = \left(\frac{3}{8}I_0\right) \times \left(\frac{1}{2}\right)^2 = \frac{3}{8}I_0 \times \frac{1}{4} = \frac{3}{32}I_0$$

11. Common Pitfalls & Exam Checklist

Exam Pitfall Alert 1: Confusing Interference and Diffraction Minima Conditions

Do not swap the conditions!

- In **Interference (YDSE)**, dark fringes occur at: $d \sin\theta = \left(n + \frac{1}{2}\right)\lambda$.

BOSONANCE

- In **Single-Slit Diffraction**, dark fringes occur at: $a \sin\theta = n\lambda$.

Remember that in single-slit diffraction, the path difference between the top and bottom of the entire slit is $a \sin\theta = \lambda$, which divides the slit into two canceling halves, creating a *minimum*, not a maximum.

Exam Pitfall Alert 2: Unpolarized vs. Polarized Light on the First Polaroid

Students frequently apply $I = I_0 \cos^2\theta$ to the *first* polaroid encountering unpolarized light. That is incorrect! For completely unpolarized light entering a polaroid, the transmitted intensity is **always exactly half** ($I = I_{\text{in}}/2$), regardless of polaroid rotation angle. Malus' Law applies only from the *second* polaroid onward.

Exam Pitfall Alert 3: Frequency vs. Speed in Denser Media

A common misconception is that a wave slows down in glass because it "loses energy." **Speed is a medium-dependent property ($v = 1/\sqrt{\mu\varepsilon}$), not an indicator of wave energy loss!** Wave energy is proportional to amplitude squared (A^2), and frequency ν is invariant.

Rapid Revision Checklist:

- ✓ Wavefront is always perpendicular to rays; spherical for point sources, plane at large distances.
- ✓ Snell's Law derivation: $\frac{\sin i}{\sin r} = \frac{v_1}{v_2} = \frac{n_2}{n_1}$. Denser medium implies slower speed.
- ✓ Phase difference $\phi = \frac{2\pi}{\lambda}\Delta x$. Resultant intensity: $I = 4I_0 \cos^2(\phi/2)$.
- ✓ YDSE fringe width: $\beta = \frac{\lambda D}{d}$; angular fringe width $\theta_0 = \frac{\lambda}{d}$.
- ✓ Submerged YDSE fringe width shrinks to β/n.
- ✓ Single-slit diffraction minima: $a \sin\theta = n\lambda$; central maximum width = $\frac{2\lambda D}{a}$.
- ✓ Polarisation occurs only for transverse waves; Malus' Law: $I = I_0 \cos^2\theta$.
- ✓ Two crossed polaroids transmit zero intensity; inserting a third polaroid at angle θ restores transmission: $I = \frac{I_0}{4}\sin^2(2\theta)$.

12. Chapter Summary & Core Formula Sheet

Concept / Law	Governing Formula	Key Variables & NEET Takeaway
Refractive Index & Speed	$n = \frac{c}{v} = \frac{\lambda_0}{\lambda}$	$\nu = \text{constant}$, v and λ decrease by factor n
Critical Angle (TIR)	$\sin i_c = \frac{n_2}{n_1} = \frac{1}{n}$	Occurs only when propagating from denser to rarer medium
Resultant Intensity	$I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\phi$	$I_{\max} = (\sqrt{I_1} + \sqrt{I_2})^2$; $I_{\min} = (\sqrt{I_1} - \sqrt{I_2})^2$
Coherent Identical Sources	$I = 4I_0 \cos^2(\phi/2)$	Incoherent addition yields average $I = 2I_0$
YDSE Fringe Width	$\beta = \frac{\lambda D}{d}$	Equally spaced; independent of fringe order n
Central Diffraction Maximum	$2\theta_1 = \frac{2\lambda}{a}, \quad \beta_0 = \frac{2\lambda D}{a}$	Twice as wide as any secondary maximum
Diffraction Minima	$a \sin\theta = n\lambda$	$n = \pm 1, \pm 2, \ldots$ (Non-zero integers)
Malus' Law	$I = I_0 \cos^2\theta$	I_0 is intensity after first polarizer ($I_{\text{in}}/2$)

BOSONANCE